

Conditional Probability

40 problems

17 textbook exercises · 14 strategic practice · 9 homework

Questions below; full solutions at the end of the chapter.

TEXTBOOK EXERCISES

Conditioning on evidence

Q1 Book Ch.2, Exercise 1

[solution · p. 45](#)

1. Ⓢ A spam filter is designed by looking at commonly occurring phrases in spam. Suppose that 80% of email is spam. In 10% of the spam emails, the phrase “free money” is used, whereas this phrase is only used in 1% of non-spam emails. A new email has just arrived, which does mention “free money”. What is the probability that it is spam?

Q2 Book Ch.2, Exercise 2

[solution · p. 45](#)

2. Ⓢ A woman is pregnant with twin boys. Twins may be either identical or fraternal (non-identical). In general, $1/3$ of twins born are identical. Obviously, identical twins must be of the same sex; fraternal twins may or may not be. Assume that identical twins are equally likely to be both boys or both girls, while for fraternal twins all possibilities are equally likely. Given the above information, what is the probability that the woman’s twins are identical?

Q3 Book Ch.2, Exercise 22

[solution · p. 45](#)

22. Ⓢ A bag contains one marble which is either green or blue, with equal probabilities. A green marble is put in the bag (so there are 2 marbles now), and then a random marble is taken out. The marble taken out is green. What is the probability that the remaining marble is also green?

23. ⑤ Let G be the event that a certain individual is guilty of a certain robbery. In gathering evidence, it is learned that an event E_1 occurred, and a little later it is also learned that another event E_2 also occurred. Is it possible that individually, these pieces of evidence increase the chance of guilt (so $P(G|E_1) > P(G)$ and $P(G|E_2) > P(G)$), but together they decrease the chance of guilt (so $P(G|E_1, E_2) < P(G)$)?

25. ⑤ A crime is committed by one of two suspects, A and B . Initially, there is equal evidence against both of them. In further investigation at the crime scene, it is found that the guilty party had a blood type found in 10% of the population. Suspect A does match this blood type, whereas the blood type of Suspect B is unknown.
- (a) Given this new information, what is the probability that A is the guilty party?
- (b) Given this new information, what is the probability that B 's blood type matches that found at the crime scene?

26. ⑤ To battle against spam, Bob installs two anti-spam programs. An email arrives, which is either legitimate (event L) or spam (event L^c), and which program j marks as legitimate (event M_j) or marks as spam (event M_j^c) for $j \in \{1, 2\}$. Assume that 10% of Bob's email is legitimate and that the two programs are each "90% accurate" in the sense that $P(M_j|L) = P(M_j^c|L^c) = 9/10$. Also assume that given whether an email is spam, the two programs' outputs are conditionally independent.
- (a) Find the probability that the email is legitimate, given that the 1st program marks it as legitimate (simplify).
- (b) Find the probability that the email is legitimate, given that both programs mark it as legitimate (simplify).
- (c) Bob runs the 1st program and M_1 occurs. He updates his probabilities and then runs the 2nd program. Let $\tilde{P}(A) = P(A|M_1)$ be the updated probability function after running the 1st program. Explain briefly in words whether or not $\tilde{P}(L|M_2) = P(L|M_1 \cap M_2)$: is conditioning on $M_1 \cap M_2$ in one step equivalent to first conditioning on M_1 , then updating probabilities, and then conditioning on M_2 ?

Independence and conditional independence

30. ⑤ A family has 3 children, creatively named A , B , and C .
- (a) Discuss intuitively (but clearly) whether the event " A is older than B " is independent of the event " A is older than C ".
- (b) Find the probability that A is older than B , given that A is older than C .

31. (S) Is it possible that an event is independent of itself? If so, when is this the case?

32. (S) Consider four nonstandard dice (the *Efron dice*), whose sides are labeled as follows (the 6 sides on each die are equally likely).

A: 4, 4, 4, 4, 0, 0

B: 3, 3, 3, 3, 3, 3

C: 6, 6, 2, 2, 2, 2

D: 5, 5, 5, 1, 1, 1

These four dice are each rolled once. Let A be the result for die A, B be the result for die B, etc.

- (a) Find $P(A > B)$, $P(B > C)$, $P(C > D)$, and $P(D > A)$.
- (b) Is the event $A > B$ independent of the event $B > C$? Is the event $B > C$ independent of the event $C > D$? Explain.

35. (S) You are going to play 2 games of chess with an opponent whom you have never played against before (for the sake of this problem). Your opponent is equally likely to be a beginner, intermediate, or a master. Depending on which, your chances of winning an individual game are 90%, 50%, or 30%, respectively.
- (a) What is your probability of winning the first game?
- (b) Congratulations: you won the first game! Given this information, what is the probability that you will also win the second game (assume that, given the skill level of your opponent, the outcomes of the games are independent)?
- (c) Explain the distinction between assuming that the outcomes of the games are independent and assuming that they are conditionally independent given the opponent's skill level. Which of these assumptions seems more reasonable, and why?

Monty Hall

38. (S) (a) Consider the following 7-door version of the Monty Hall problem. There are 7 doors, behind one of which there is a car (which you want), and behind the rest of which there are goats (which you don't want). Initially, all possibilities are equally likely for where the car is. You choose a door. Monty Hall then opens 3 goat doors, and offers you the option of switching to any of the remaining 3 doors.
- Assume that Monty Hall knows which door has the car, will always open 3 goat doors and offer the option of switching, and that Monty chooses with equal probabilities from all his choices of which goat doors to open. Should you switch? What is your probability of success if you switch to one of the remaining 3 doors?
- (b) Generalize the above to a Monty Hall problem where there are $n \geq 3$ doors, of which Monty opens m goat doors, with $1 \leq m \leq n - 2$.

39. (S) Consider the Monty Hall problem, except that Monty enjoys opening door 2 more than he enjoys opening door 3, and if he has a choice between opening these two doors, he opens door 2 with probability p , where $\frac{1}{2} \leq p \leq 1$.

To recap: there are three doors, behind one of which there is a car (which you want), and behind the other two of which there are goats (which you don't want). Initially, all possibilities are equally likely for where the car is. You choose a door, which for concreteness we assume is door 1. Monty Hall then opens a door to reveal a goat, and offers you the option of switching. Assume that Monty Hall knows which door has the car, will always open a goat door and offer the option of switching, and as above assume that if Monty Hall has a choice between opening door 2 and door 3, he chooses door 2 with probability p (with $\frac{1}{2} \leq p \leq 1$).

(a) Find the unconditional probability that the strategy of always switching succeeds (unconditional in the sense that we do not condition on which of doors 2 or 3 Monty opens).

(b) Find the probability that the strategy of always switching succeeds, given that Monty opens door 2.

(c) Find the probability that the strategy of always switching succeeds, given that Monty opens door 3.

First-step analysis and gambler's ruin

42. (S) A fair die is rolled repeatedly, and a running total is kept (which is, at each time, the total of all the rolls up until that time). Let p_n be the probability that the running total is ever *exactly* n (assume the die will always be rolled enough times so that the running total will eventually exceed n , but it may or may not ever equal n).

(a) Write down a recursive equation for p_n (relating p_n to earlier terms p_k in a simple way). Your equation should be true for all positive integers n , so give a definition of p_0 and p_k for $k < 0$ so that the recursive equation is true for small values of n .

(b) Find p_7 .

(c) Give an intuitive explanation for the fact that $p_n \rightarrow 1/3.5 = 2/7$ as $n \rightarrow \infty$.

44. (S) Calvin and Hobbes play a match consisting of a series of games, where Calvin has probability p of winning each game (independently). They play with a "win by two" rule: the first player to win two games more than his opponent wins the match. Find the probability that Calvin wins the match (in terms of p), in two different ways:

(a) by conditioning, using the law of total probability.

(b) by interpreting the problem as a gambler's ruin problem.

Simpson's paradox

49. (S) (a) Is it possible to have events A, B, C such that $P(A|C) < P(B|C)$ and $P(A|C^c) < P(B|C^c)$, yet $P(A) > P(B)$? That is, A is less likely than B given that C is true, and also less likely than B given that C is false, yet A is more likely than B if we're given no information about C . Show this is impossible (with a short proof) or find a counterexample (with a story interpreting A, B, C).
- (b) If the scenario in (a) is possible, is it a special case of Simpson's paradox, equivalent to Simpson's paradox, or neither? If it is impossible, explain intuitively why it is impossible even though Simpson's paradox is possible.

50. (S) Consider the following conversation from an episode of *The Simpsons*:
- Lisa: Dad, I think he's an ivory dealer! His boots are ivory, his hat is ivory, and I'm pretty sure that check is ivory.
- Homer: Lisa, a guy who has lots of ivory is less likely to hurt Stampy than a guy whose ivory supplies are low.

Here Homer and Lisa are debating the question of whether or not the man (named Blackheart) is likely to hurt Stampy the Elephant if they sell Stampy to him. They clearly disagree about how to use their observations about Blackheart to learn about the probability (conditional on the evidence) that Blackheart will hurt Stampy.

- (a) Define clear notation for the various events of interest here.
- (b) Express Lisa's and Homer's arguments (Lisa's is partly implicit) as conditional probability statements in terms of your notation from (a).
- (c) Assume it is true that someone who has a lot of a commodity will have less desire to acquire more of the commodity. Explain what is wrong with Homer's reasoning that the evidence about Blackheart makes it less likely that he will harm Stampy.

53. (S) The book *Red State, Blue State, Rich State, Poor State* by Andrew Gelman [13] discusses the following election phenomenon: within any U.S. state, a wealthy voter is more likely to vote for a Republican than a poor voter, yet the wealthier states tend to favor Democratic candidates! In short: rich individuals (in any state) tend to vote for Republicans, while states with a higher percentage of rich people tend to favor Democrats.
- (a) Assume for simplicity that there are only 2 states (called Red and Blue), each of which has 100 people, and that each person is either rich or poor, and either a Democrat or a Republican. Make up numbers consistent with the above, showing how this phenomenon is possible, by giving a 2×2 table for each state (listing how many people in each state are rich Democrats, etc.).
- (b) In the setup of (a) (not necessarily with the numbers you made up there), let D be the event that a randomly chosen person is a Democrat (with all 200 people equally likely), and B be the event that the person lives in the Blue State. Suppose that 10 people move from the Blue State to the Red State. Write P_{old} and P_{new} for probabilities before and after they move. Assume that people do not change parties, so we have $P_{\text{new}}(D) = P_{\text{old}}(D)$. Is it possible that *both* $P_{\text{new}}(D|B) > P_{\text{old}}(D|B)$ and $P_{\text{new}}(D|B^c) > P_{\text{old}}(D|B^c)$ are true? If so, explain how it is possible and why it does not contradict the law of total probability $P(D) = P(D|B)P(B) + P(D|B^c)P(B^c)$; if not, show that it is impossible.

STRATEGIC PRACTICE

2 Independence

Q18 SP 2 §2 · Problem 1

[solution · p. 52](#)

1. Is it possible that an event is independent of itself? If so, when?

Q19 SP 2 §2 · Problem 2

[solution · p. 52](#)

2. Is it always true that if A and B are independent events, then A^c and B^c are independent events? Show that it is, or give a counterexample.

Q20 SP 2 §2 · Problem 3

[solution · p. 53](#)

3. Give an example of 3 events A, B, C which are pairwise independent but not independent. Hint: find an example where whether C occurs is completely determined if we know whether A occurred and whether B occurred, but completely undetermined if we know only one of these things.

Q21 SP 2 §2 · Problem 4

[solution · p. 53](#)

4. Give an example of 3 events A, B, C which are not independent, yet satisfy $P(A \cap B \cap C) = P(A)P(B)P(C)$. Hint: consider simple and extreme cases.

3 Thinking Conditionally

Q22 SP 2 §3 · Problem 1

[solution · p. 53](#)

1. A bag contains one marble which is either green or blue, with equal probabilities. A green marble is put in the bag (so there are 2 marbles now), and then a random marble is taken out. The marble taken out is green. What is the probability that the remaining marble is also green?

Historical note: this problem was first posed by Lewis Carroll in 1893.

Q23 SP 2 §3 · Problem 2

[solution · p. 54](#)

2. A spam filter is designed by looking at commonly occurring phrases in spam. Suppose that 80% of email is spam. In 10% of the spam emails, the phrase “free money” is used, whereas this phrase is only used in 1% of non-spam emails. A new email has just arrived, which does mention “free money”. What is the probability that it is spam?

3. Let G be the event that a certain individual is guilty of a certain robbery. In gathering evidence, it is learned that an event E_1 occurred, and a little later it is also learned that another event E_2 also occurred.
- (a) Is it possible that individually, these pieces of evidence increase the chance of guilt (so $P(G|E_1) > P(G)$ and $P(G|E_2) > P(G)$), but together they decrease the chance of guilt (so $P(G|E_1, E_2) < P(G)$)?
- (b) Show that the probability of guilt given the evidence is the same regardless of whether we update our probabilities all at once, or in two steps (after getting the first piece of evidence, and again after getting the second piece of evidence). That is, we can either update all at once (computing $P(G|E_1, E_2)$ in one step), or we can first update based on E_1 , so that our new probability function is $P_{\text{new}}(A) = P(A|E_1)$, and then update based on E_2 by computing $P_{\text{new}}(G|E_2)$.

4. A crime is committed by one of two suspects, A and B . Initially, there is equal evidence against both of them. In further investigation at the crime scene, it is found that the guilty party had a blood type found in 10% of the population. Suspect A does match this blood type, whereas the blood type of Suspect B is unknown.
- (a) Given this new information, what is the probability that A is the guilty party?
- (b) Given this new information, what is the probability that B 's blood type matches that found at the crime scene?

5. You are going to play 2 games of chess with an opponent whom you have never played against before (for the sake of this problem). Your opponent is equally likely to be a beginner, intermediate, or a master. Depending on which, your chances of winning an individual game are 90%, 50%, or 30%, respectively.
- (a) What is your probability of winning the first game?
- (b) Congratulations: you won the first game! Given this information, what is the probability that you will also win the second game (assume that, given the skill level of your opponent, the outcomes of the games are independent)?
- (c) Explain the distinction between assuming that the outcomes of the games are independent and assuming that they are conditionally independent given the opponent's skill level. Which of these assumptions seems more reasonable, and why?

1. Consider the Monty Hall problem, except that Monty enjoys opening Door 2 more than he enjoys opening Door 3, and if he has a choice between opening these two doors, he opens Door 2 with probability p , where $\frac{1}{2} \leq p \leq 1$.

To recap: there are three doors, behind one of which there is a car (which you want), and behind the other two of which there are goats (which you don't want). Initially, all possibilities are equally likely for where the car is. You choose a door, which for concreteness we assume is Door 1. Monty Hall then opens a door to reveal a goat, and offers you the option of switching. Assume that Monty Hall knows which door has the car, will always open a goat door and offer the option of switching, and as above assume that if Monty Hall has a choice between opening Door 2 and Door 3, he chooses Door 2 with probability p (with $\frac{1}{2} \leq p \leq 1$).

- (a) Find the unconditional probability that the strategy of always switching succeeds (unconditional in the sense that we do not condition on which of Doors 2,3 Monty opens).
- (b) Find the probability that the strategy of always switching succeeds, given that Monty opens Door 2.
- (c) Find the probability that the strategy of always switching succeeds, given that Monty opens Door 3.

2. For each statement below, either show that it is true or give a counterexample. Throughout, X, Y, Z are discrete random variables.
 - (a) If X and Y are independent and Y and Z are independent, then X and Z are independent.
 - (b) If X and Y are independent, then they are conditionally independent given Z .
 - (c) If X and Y are conditionally independent given Z , then they are independent.
 - (d) If X and Y have the same distribution given Z , i.e., for all a and z , we have $P(X = a|Z = z) = P(Y = a|Z = z)$, then X and Y have the same distribution.

1. (a) Is it possible to have events A, B, E such that $P(A|E) < P(B|E)$ and $P(A|E^c) < P(B|E^c)$, yet $P(A) > P(B)$? That is, A is less likely under B given that E is true, and also given that E is false, yet A is more likely than B if given no information about E . Show this is impossible (with a short proof) or find a counterexample (with a “story” interpreting A, B, E).
- (b) Is it possible to have events A, B, E such that $P(A|B, E) < P(A|B^c, E)$ and $P(A|B, E^c) < P(A|B^c, E^c)$, yet $P(A|B) > P(A|B^c)$? That is, given that E is true, learning B is evidence against A , and similarly given that E is false; but given no information about E , learning that B is true is evidence in favor of A . Show this is impossible (with a short proof) or find a counterexample (with a “story” interpreting A, B, E).

2. Consider the following conversation from an episode of *The Simpsons*:

Lisa: *Dad, I think he's an ivory dealer! His boots are ivory, his hat is ivory, and I'm pretty sure that check is ivory.*

Homer: *Lisa, a guy who's got lots of ivory is less likely to hurt Stampy than a guy whose ivory supplies are low.*

Here Homer and Lisa are debating the question of whether or not the man (named Blackheart) is likely to hurt Stampy the Elephant if they sell Stampy to him. They clearly disagree about how to use their observations about Blackheart to learn about the probability (conditional on the evidence) that Blackheart will hurt Stampy.

- (a) Define clear notation for the various events of interest here.
- (b) Express Lisa's and Homer's arguments (Lisa's is partly implicit) as conditional probability statements in terms of your notation from (a).
- (c) Assume it is true that someone who has a lot of a commodity will have less desire to acquire more of the commodity. Explain what is wrong with Homer's reasoning that the evidence about Blackheart makes it less likely that he will harm Stampy.

3 Gambler's Ruin

Q31 SP 3 §3 · Problem 1

[solution · p. 61](#)

1. A gambler repeatedly plays a game where in each round, he wins a dollar with probability $1/3$ and loses a dollar with probability $2/3$. His strategy is “quit when he is ahead by \$2,” though some suspect he is a gambling addict anyway. Suppose that he starts with a million dollars. Show that the probability that he’ll ever be ahead by \$2 is less than $1/4$.

HOMEWORK

Q32 HW 2 · Problem 3

[solution · p. 62](#)

3. A family has 3 children, creatively named A , B , and C .
 - (a) Discuss intuitively (but clearly) whether the event “ A is older than B ” is independent of the event “ A is older than C .”
 - (b) Find the probability that A is older than B , given that A is older than C .

Q33 HW 2 · Problem 4

[solution · p. 62](#)

4. Two coins are in a hat. The coins look alike, but one coin is fair (with probability $1/2$ of Heads), while the other coin is biased, with probability $1/4$ of Heads. One of the coins is randomly pulled from the hat, without knowing which of the two it is. Call the chosen coin “Coin C ”.
 - (a) Coin C is tossed twice, showing Heads both times. Given this information, what is the probability that Coin C is the fair coin?
 - (b) Are the events “first toss of Coin C is Heads” and “second toss of Coin C is Heads” independent? Explain.
 - (c) Find the probability that in 10 flips of Coin C , there will be exactly 3 Heads. (The coin is equally likely to be either of the 2 coins; do not assume it already landed Heads twice as in (a).)

Q34 HW 2 · Problem 5

[solution · p. 63](#)

5. A woman has been murdered, and her husband is accused of having committed the murder. It is known that the man abused his wife repeatedly in the past, and the prosecution argues that this is important evidence pointing towards the man's guilt. The defense attorney says that the history of abuse is irrelevant, as only 1 in 1000 men who beat their wives end up murdering them.

Assume that the defense attorney's 1 in 1000 figure is correct, and that half of men who murder their wives previously abused them. Also assume that 20% of murdered women were killed by their husbands, and that if a woman is murdered and the husband is *not* guilty, then there is only a 10% chance that the husband abused her. What is the probability that the man is guilty? Is the prosecution right that the abuse is important evidence in favor of guilt?

Q35 HW 2 · Problem 6

[solution · p. 64](#)

6. A family has two children. Assume that birth month is independent of gender, with boys and girls equally likely and all months equally likely, and assume that the elder child's characteristics are independent of the younger child's characteristics).

(a) Find the probability that both are girls, given that the elder child is a girl who was born in March.

(b) Find the probability that both are girls, given that at least one is a girl who was born in March.

Q36 HW 3 · Problem 1

[solution · p. 65](#)

1. (a) Consider the following 7-door version of the Monty Hall problem. There are 7 doors, behind one of which there is a car (which you want), and behind the rest of which there are goats (which you don't want). Initially, all possibilities are equally likely for where the car is. You choose a door. Monty Hall then opens 3 goat doors, and offers you the option of switching to any of the remaining 3 doors.

Assume that Monty Hall knows which door has the car, will always open 3 goat doors and offer the option of switching, and that Monty chooses with equal probabilities from all his choices of which goat doors to open. Should you switch? What is your probability of success if you switch to one of the remaining 3 doors?

(b) Generalize the above to a Monty Hall problem where there are $n \geq 3$ doors, of which Monty opens m goat doors, with $1 \leq m \leq n - 2$.

Q37 HW 3 · Problem 2

[solution · p. 66](#)

2. The *odds* of an event with probability p are defined to be $\frac{p}{1-p}$, e.g., an event with probability $3/4$ is said to have odds of 3 to 1 in favor (or 1 to 3 against). We are interested in a hypothesis H (which we think of as a event), and we gather new data as evidence (expressed as an event D) to study the hypothesis. The *prior* probability of H is our probability for H being true before we gather the new data; the *posterior* probability of H is our probability for it after we gather the new data. The *likelihood ratio* is defined as $\frac{P(D|H)}{P(D|H^c)}$.

(a) Show that Bayes' rule can be expressed in terms of odds as follows: *the posterior odds of a hypothesis H are the prior odds of H times the likelihood ratio.*

(b) As in the example from class, suppose that a patient tests positive for a disease afflicting 1% of the population. For a patient who has the disease, there is a 95% chance of testing positive (in medical statistics, this is called the *sensitivity* of the test); for a patient who doesn't have the disease, there is a 95% chance of testing negative test (in medical statistics, this is called the *specificity* of the test).

The patient gets a second, independent test done (with the same sensitivity and specificity), and again tests positive. Use the odds form of Bayes' rule to find the probability that the patient has the disease, given the evidence, *in two ways*: in one step, conditioning on both test results simultaneously, and in two steps, first updating the probabilities based on the first test result, and then updating again based on the second test result.

Q38 HW 3 · Problem 3

[solution · p. 68](#)

3. Is it possible to have events A_1, A_2, B, C with $P(A_1|B) > P(A_1|C)$ and $P(A_2|B) > P(A_2|C)$, yet $P(A_1 \cup A_2|B) < P(A_1 \cup A_2|C)$? If so, find an example (with a “story”

interpreting the events, as well as giving specific numbers); otherwise, show that it is impossible for this phenomenon to happen.

Q39 HW 3 · Problem 4

[solution · p. 69](#)

4. Calvin and Hobbes play a match consisting of a series of games, where Calvin has probability p of winning each game (independently). They play with a “win by two” rule: the first player to win two games more than his opponent wins the match. Find the probability that Calvin wins the match (in terms of p), in two different ways:

- (a) by conditioning, using the law of total probability.
- (b) by interpreting the problem as a gambler's ruin problem.

Q40 HW 3 · Problem 5

[solution · p. 69](#)

5. A fair die is rolled repeatedly, and a running total is kept (which is, at each time, the total of all the rolls up until that time). Let p_n be the probability that the running total is ever *exactly* n (assume the die will always be rolled enough times so that the running total will eventually exceed n , but it may or may not ever equal n).
- (a) Write down a recursive equation for p_n (relating p_n to earlier terms p_k in a simple way). Your equation should be true for all positive integers n , so give a definition of p_0 and p_k for $k < 0$ so that the recursive equation is true for small values of n .
- (b) Find p_7 .
- (c) Give an intuitive explanation for the fact that $p_n \rightarrow 1/3.5 = 2/7$ as $n \rightarrow \infty$.

CHAPTER 2 · SOLUTIONS

S1 Book Ch.2, Exercise 1

[question · p. 32](#)

Solution: Let S be the event that an email is spam and F be the event that an email has the “free money” phrase. By Bayes’ rule,

$$P(S|F) = \frac{P(F|S)P(S)}{P(F)} = \frac{0.1 \cdot 0.8}{0.1 \cdot 0.8 + 0.01 \cdot 0.2} = \frac{80/1000}{82/1000} = \frac{80}{82} \approx 0.9756.$$

S2 Book Ch.2, Exercise 2

[question · p. 32](#)

Solution: By Bayes’ rule,

$$P(\text{identical}|BB) = \frac{P(BB|\text{identical})P(\text{identical})}{P(BB)} = \frac{\frac{1}{2} \cdot \frac{1}{3}}{\frac{1}{2} \cdot \frac{1}{3} + \frac{1}{4} \cdot \frac{2}{3}} = 1/2.$$

S3 Book Ch.2, Exercise 22

[question · p. 32](#)

Solution: Let A be the event that the initial marble is green, B be the event that the removed marble is green, and C be the event that the remaining marble is green. We need to find $P(C|B)$. There are several ways to find this; one natural way is to condition on whether the initial marble is green:

$$P(C|B) = P(C|B, A)P(A|B) + P(C|B, A^c)P(A^c|B) = 1P(A|B) + 0P(A^c|B).$$

To find $P(A|B)$, use Bayes’ Rule:

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)} = \frac{1/2}{P(B|A)P(A) + P(B|A^c)P(A^c)} = \frac{1/2}{1/2 + 1/4} = \frac{2}{3}.$$

So $P(C|B) = 2/3$.

Historical note: This problem was first posed by Lewis Carroll in 1893.

S4 Book Ch.2, Exercise 23

[question · p. 33](#)

Solution: Yes, this is possible. In fact, it is possible to have two events which separately provide evidence in favor of G , yet which together preclude G ! For example, suppose that the crime was committed between 1 pm and 3 pm on a certain day. Let E_1 be the event that the suspect was at a specific nearby coffeeshop from 1 pm to 2 pm that day, and let E_2 be the event that the suspect was at the nearby coffeeshop from 2 pm to 3 pm that day. Then $P(G|E_1) > P(G)$, $P(G|E_2) > P(G)$ (assuming that being in the vicinity helps show that the suspect had the opportunity to commit the crime), yet $P(G|E_1 \cap E_2) < P(G)$ (as being in the coffeehouse from 1 pm to 3 pm gives the suspect an alibi for the full time).

S5 Book Ch.2, Exercise 25

[question · p. 33](#)

Solution:

(a) Let M be the event that A 's blood type matches the guilty party's and for brevity, write A for “ A is guilty” and B for “ B is guilty”. By Bayes' Rule,

$$P(A|M) = \frac{P(M|A)P(A)}{P(M|A)P(A) + P(M|B)P(B)} = \frac{1/2}{1/2 + (1/10)(1/2)} = \frac{10}{11}.$$

(We have $P(M|B) = 1/10$ since, given that B is guilty, the probability that A 's blood type matches the guilty party's is the same probability as for the general population.)

(b) Let C be the event that B 's blood type matches, and condition on whether B is guilty. This gives

$$P(C|M) = P(C|M, A)P(A|M) + P(C|M, B)P(B|M) = \frac{1}{10} \cdot \frac{10}{11} + \frac{1}{11} = \frac{2}{11}.$$

S6 Book Ch.2, Exercise 26

question · p. 33

Solution:

(a) By Bayes' rule,

$$P(L|M_1) = \frac{P(M_1|L)P(L)}{P(M_1)} = \frac{\frac{9}{10} \cdot \frac{1}{10}}{\frac{9}{10} \cdot \frac{1}{10} + \frac{1}{10} \cdot \frac{9}{10}} = \frac{1}{2}.$$

(b) By Bayes' rule,

$$P(L|M_1, M_2) = \frac{P(M_1, M_2|L)P(L)}{P(M_1, M_2)} = \frac{(\frac{9}{10})^2 \cdot \frac{1}{10}}{(\frac{9}{10})^2 \cdot \frac{1}{10} + (\frac{1}{10})^2 \cdot \frac{9}{10}} = \frac{9}{10}.$$

(c) Yes, they are the same, since Bayes' rule is coherent. The probability of an event given various pieces of evidence does not depend on the order in which the pieces of evidence are incorporated into the updated probabilities.

S7 Book Ch.2, Exercise 30

question · p. 33

Solution:

(a) They are not independent: knowing that A is older than B makes it more likely that A is older than C , as the if A is older than B , then the only way that A can be younger than C is if the birth order is CAB , whereas the birth orders ABC and ACB are both compatible with A being older than B . To make this more intuitive, think of an extreme case where there are 100 children instead of 3, call them $A_1, \dots, A_{100}$. Given that A_1 is older than all of $A_2, A_3, \dots, A_{99}$, it's clear that A_1 is very old (relatively), whereas there isn't evidence about where A_{100} fits into the birth order.

(b) Writing $x > y$ to mean that x is older than y ,

$$P(A > B | A > C) = \frac{P(A > B, A > C)}{P(A > C)} = \frac{1/3}{1/2} = \frac{2}{3}$$

since $P(A > B, A > C) = P(A \text{ is the eldest child}) = 1/3$ (unconditionally, any of the 3 children is equally likely to be the eldest).

Solution: Let A be an event. If A is independent of itself, then $P(A) = P(A \cap A) = P(A)^2$, so $P(A)$ is 0 or 1. So this is only possible in the extreme cases that the event has probability 0 or 1.

Solution:

(a)

$$P(A > B) = P(A = 4) = 2/3$$

$$P(B > C) = P(C = 2) = 2/3$$

$$P(C > D) = P(C = 6) + P(C = 2, D = 1) = 2/3$$

$$P(D > A) = P(D = 5) + P(D = 1, A = 0) = 2/3$$

(b) The event $A > B$ is independent of the event $B > C$ since $A > B$ is the same thing as $A = 4$, knowledge of which gives no information about $B > C$ (which is the same thing as $C = 2$). On the other hand, $B > C$ is *not* independent of $C > D$ since $P(C > D | C = 2) = 1/2 \neq 1 = P(C > D | C \neq 2)$.

Solution:

(a) Let W_i be the event of winning the i th game. By the law of total probability,

$$P(W_1) = (0.9 + 0.5 + 0.3)/3 = 17/30.$$

(b) We have $P(W_2 | W_1) = P(W_2, W_1) / P(W_1)$. The denominator is known from (a), while the numerator can be found by conditioning on the skill level of the opponent:

$$P(W_1, W_2) = \frac{1}{3}P(W_1, W_2 | \text{beginner}) + \frac{1}{3}P(W_1, W_2 | \text{intermediate}) + \frac{1}{3}P(W_1, W_2 | \text{expert}).$$

Since W_1 and W_2 are conditionally independent given the skill level of the opponent, this becomes

$$P(W_1, W_2) = (0.9^2 + 0.5^2 + 0.3^2)/3 = 23/60.$$

So

$$P(W_2 | W_1) = \frac{23/60}{17/30} = 23/34.$$

(c) Independence here means that knowing one game's outcome gives no information about the other game's outcome, while conditional independence is the same statement where all probabilities are conditional on the opponent's skill level. Conditional independence given the opponent's skill level is a more reasonable assumption here. This is because winning the first game gives information about the opponent's skill level, which in turn gives information about the result of the second game.

That is, if the opponent's skill level is treated as fixed and known, then it may be reasonable to assume independence of games given this information; with the opponent's skill level random, earlier games can be used to help infer the opponent's skill level, which affects the probabilities for future games.

Solution:

(a) Assume the doors are labeled such that you choose door 1 (to simplify notation), and suppose first that you follow the “stick to your original choice” strategy. Let S be the event of success in getting the car, and let C_j be the event that the car is behind door j . Conditioning on which door has the car, we have

$$P(S) = P(S|C_1)P(C_1) + \cdots + P(S|C_7)P(C_7) = P(C_1) = \frac{1}{7}.$$

Let M_{ijk} be the event that Monty opens doors i, j, k . Then

$$P(S) = \sum_{i,j,k} P(S|M_{ijk})P(M_{ijk})$$

(summed over all i, j, k with $2 \leq i < j < k \leq 7$.) By symmetry, this gives

$$P(S|M_{ijk}) = P(S) = \frac{1}{7}$$

for all i, j, k with $2 \leq i < j < k \leq 7$. Thus, the conditional probability that the car is behind 1 of the remaining 3 doors is $6/7$, which gives $2/7$ for each. So you should switch, thus making your probability of success $2/7$ rather than $1/7$.

(b) By the same reasoning, the probability of success for “stick to your original choice” is $\frac{1}{n}$, both unconditionally and conditionally. Each of the $n - m - 1$ remaining doors has conditional probability $\frac{n-1}{(n-m-1)n}$ of having the car. This value is greater than $\frac{1}{n}$, so you should switch, thus obtaining probability $\frac{n-1}{(n-m-1)n}$ of success (both conditionally and unconditionally).

Solution:

(a) Let C_j be the event that the car is hidden behind door j and let W be the event that we win using the switching strategy. Using the law of total probability, we can find the unconditional probability of winning:

$$\begin{aligned} P(W) &= P(W|C_1)P(C_1) + P(W|C_2)P(C_2) + P(W|C_3)P(C_3) \\ &= 0 \cdot 1/3 + 1 \cdot 1/3 + 1 \cdot 1/3 = 2/3. \end{aligned}$$

(b) A tree method works well here (delete the paths which are no longer relevant after the conditioning, and reweight the remaining values by dividing by their sum), or we can use Bayes' rule and the law of total probability (as below).

Let D_i be the event that Monty opens Door i . Note that we are looking for $P(W|D_2)$, which is the same as $P(C_3|D_2)$ as we first choose Door 1 and then switch to Door 3. By Bayes' rule and the law of total probability,

$$\begin{aligned} P(C_3|D_2) &= \frac{P(D_2|C_3)P(C_3)}{P(D_2)} \\ &= \frac{P(D_2|C_3)P(C_3)}{P(D_2|C_1)P(C_1) + P(D_2|C_2)P(C_2) + P(D_2|C_3)P(C_3)} \\ &= \frac{1 \cdot 1/3}{p \cdot 1/3 + 0 \cdot 1/3 + 1 \cdot 1/3} \\ &= \frac{1}{1+p}. \end{aligned}$$

(c) The structure of the problem is the same as part (b) (except for the condition that $p \geq 1/2$, which was not needed above). Imagine repainting doors 2 and 3, reversing which is called which. By part (b) with $1-p$ in place of p , $P(C_2|D_3) = \frac{1}{1+(1-p)} = \frac{1}{2-p}$.

Solution:

(a) We will find something to condition on to reduce the case of interest to earlier, simpler cases. This is achieved by the useful strategy of *first step analysis*. Let p_n be the probability that the running total is ever *exactly* n . Note that if, for example, the first

throw is a 3, then the probability of reaching n exactly is p_{n-3} since starting from that point, we need to get a total of $n - 3$ exactly. So

$$p_n = \frac{1}{6}(p_{n-1} + p_{n-2} + p_{n-3} + p_{n-4} + p_{n-5} + p_{n-6}),$$

where we define $p_0 = 1$ (which makes sense anyway since the running total is 0 before the first toss) and $p_k = 0$ for $k < 0$.

(b) Using the recursive equation in (a), we have

$$\begin{aligned} p_1 &= \frac{1}{6}, \quad p_2 = \frac{1}{6}\left(1 + \frac{1}{6}\right), \quad p_3 = \frac{1}{6}\left(1 + \frac{1}{6}\right)^2, \\ p_4 &= \frac{1}{6}\left(1 + \frac{1}{6}\right)^3, \quad p_5 = \frac{1}{6}\left(1 + \frac{1}{6}\right)^4, \quad p_6 = \frac{1}{6}\left(1 + \frac{1}{6}\right)^5. \end{aligned}$$

Hence,

$$p_7 = \frac{1}{6}(p_1 + p_2 + p_3 + p_4 + p_5 + p_6) = \frac{1}{6} \left(\left(1 + \frac{1}{6}\right)^6 - 1 \right) \approx 0.2536.$$

(c) An intuitive explanation is as follows. The average number thrown by the die is (total of dots)/6, which is $21/6 = 7/2$, so that every throw adds on an average of $7/2$. We can therefore expect to land on 2 out of every 7 numbers, and the probability of landing on any particular number is $2/7$. A mathematical derivation (which was not requested in the problem) can be given as follows:

$$\begin{aligned} & p_{n+1} + 2p_{n+2} + 3p_{n+3} + 4p_{n+4} + 5p_{n+5} + 6p_{n+6} \\ &= p_{n+1} + 2p_{n+2} + 3p_{n+3} + 4p_{n+4} + 5p_{n+5} \\ & \quad + p_n + p_{n+1} + p_{n+2} + p_{n+3} + p_{n+4} + p_{n+5} \\ &= p_n + 2p_{n+1} + 3p_{n+2} + 4p_{n+3} + 5p_{n+4} + 6p_{n+5} \\ &= \dots \\ &= p_{-5} + 2p_{-4} + 3p_{-3} + 4p_{-2} + 5p_{-1} + 6p_0 = 6. \end{aligned}$$

Taking the limit of the lefthand side as n goes to ∞ , we have

$$(1 + 2 + 3 + 4 + 5 + 6) \lim_{n \rightarrow \infty} p_n = 6,$$

so $\lim_{n \rightarrow \infty} p_n = 2/7$.

Solution:

(a) Let C be the event that Calvin wins the match, $X \sim \text{Bin}(2, p)$ be how many of the first 2 games he wins, and $q = 1 - p$. Then

$$P(C) = P(C|X=0)q^2 + P(C|X=1)(2pq) + P(C|X=2)p^2 = 2pqP(C) + p^2,$$

so $P(C) = \frac{p^2}{1-2pq}$. This can also be written as $\frac{p^2}{p^2+q^2}$, since $p + q = 1$.

Sanity check: Note that this should (and does) reduce to 1 for $p = 1$, 0 for $p = 0$, and

$\frac{1}{2}$ for $p = \frac{1}{2}$. Also, it makes sense that the probability of Hobbes winning, which is $1 - P(C) = \frac{q^2}{p^2 + q^2}$, can also be obtained by swapping p and q .

(b) The problem can be thought of as a gambler's ruin where each player starts out with \$2. So the probability that Calvin wins the match is

$$\frac{1 - (q/p)^2}{1 - (q/p)^4} = \frac{(p^2 - q^2)/p^2}{(p^4 - q^4)/p^4} = \frac{(p^2 - q^2)/p^2}{(p^2 - q^2)(p^2 + q^2)/p^4} = \frac{p^2}{p^2 + q^2},$$

which agrees with the above.

S15 Book Ch.2, Exercise 49

question · p. 36

Solution:

(a) It is *not* possible, as seen using the law of total probability:

$$P(A) = P(A|C)P(C) + P(A|C^c)P(C^c) < P(B|C)P(C) + P(B|C^c)P(C^c) = P(B).$$

(b) In Simpson's paradox, using the notation from the chapter, we can expand out $P(A|B)$ and $P(A|B^c)$ using LOTP to condition on C , but the inequality can flip because of the weights such as $P(C|B)$ on the terms (e.g., Dr. Nick performs a lot more Band-Aid removals than Dr. Hibbert). In this problem, the weights $P(C)$ and $P(C^c)$ are the same in both expansions, so the inequality is preserved.

S16 Book Ch.2, Exercise 50

question · p. 36

Solution:

(a) Let H be the event that the man will hurt Stampy, let L be the event that a man has lots of ivory, and let D be the event that the man is an ivory dealer.

(b) Lisa observes that L is true. She suggests (reasonably) that this evidence makes D more likely, i.e., $P(D|L) > P(D)$. Implicitly, she suggests that this makes it likely that the man will hurt Stampy, i.e.,

$$P(H|L) > P(H|L^c).$$

Homer argues that

$$P(H|L) < P(H|L^c).$$

(c) Homer does not realize that observing that Blackheart has so much ivory makes it much more likely that Blackheart is an ivory dealer, which in turn makes it more likely that the man will hurt Stampy. This is an example of Simpson's paradox. It may be true that, *controlling for whether or not Blackheart is a dealer*, having high ivory supplies makes it less likely that he will harm Stampy: $P(H|L, D) < P(H|L^c, D)$ and $P(H|L, D^c) < P(H|L^c, D^c)$. However, this does not imply that $P(H|L) < P(H|L^c)$.

S17 Book Ch.2, Exercise 53

question · p. 36

Solution:

(a) Here are two tables that are as desired:

Red	Dem	Rep	Total	Blue	Dem	Rep	Total
Rich	5	25	30	Rich	45	15	60
Poor	20	50	70	Poor	35	5	40
Total	25	75	100	Total	80	20	100

In these tables, within each state a rich person is more likely to be a Republican than a poor person; but the richer state has a higher percentage of Democrats than the poorer state. Of course, there are many possible tables that work.

The above example is a form of Simpson's paradox: aggregating the two tables seems to give different conclusions than conditioning on which state a person is in. Letting D, W, B be the events that a randomly chosen person is a Democrat, wealthy, and from the Blue State (respectively), for the above numbers we have $P(D|W, B) < P(D|W^c, B)$ and $P(D|W, B^c) < P(D|W^c, B^c)$ (controlling for whether the person is in the Red State or the Blue State, a poor person is more likely to be a Democrat than a rich person),

but $P(D|W) > P(D|W^c)$ (stemming from the fact that the Blue State is richer and more Democratic).

(b) Yes, it is possible. Suppose with the numbers from (a) that 10 people move from the Blue State to the Red State, of whom 5 are Democrats and 5 are Republicans. Then $P_{\text{new}}(D|B) = 75/90 > 80/100 = P_{\text{old}}(D|B)$ and $P_{\text{new}}(D|B^c) = 30/110 > 25/100 = P_{\text{old}}(D|B^c)$. Intuitively, this makes sense since the Blue State has a higher percentage of Democrats initially than the Red State, and the people who move have a percentage of Democrats which is between these two values.

This result does not contradict the law of total probability since the weights $P(B), P(B^c)$ also change: $P_{\text{new}}(B) = 90/200$, while $P_{\text{old}}(B) = 1/2$. The phenomenon could not occur if an equal number of people also move from the Red State to the Blue State (so that $P(B)$ is kept constant).

S18 SP 2 §2 · Problem 1

question · p. 37

1. Is it possible that an event is independent of itself? If so, when?

Let A be an event. If A is independent of itself, then $P(A) = P(A \cap A) = P(A)^2$, so $P(A)$ is 0 or 1. So this is only possible in the extreme cases that the event has probability 0 or 1.

S19 SP 2 §2 · Problem 2

question · p. 37

2. Is it always true that if A and B are independent events, then A^c and B^c are independent events? Show that it is, or give a counterexample.

Yes, because we have

$$P(A^c \cap B^c) = 1 - P(A \cup B) = 1 - (P(A) + P(B) - P(A \cap B));$$

since A and B are independent, this becomes

$$1 - P(A) - P(B) + P(A)P(B) = (1 - P(A))(1 - P(B)) = P(A^c)P(B^c).$$

3. Give an example of 3 events A, B, C which are pairwise independent but not independent. Hint: find an example where whether C occurs is completely determined if we know whether A occurred and whether B occurred, but completely undetermined if we know only one of these things.

Consider two fair, independent coin tosses, and let A be the event that the first toss is Heads, B be the event that the second toss is Heads, and C be the event that the two tosses have the same result. Then A, B, C are dependent since $P(A \cap B \cap C) = P(A \cap B) = P(A)P(B) = 1/4 \neq 1/8 = P(A)P(B)P(C)$, but they are pairwise independent: A and B are independent by definition; A and C are independent since $P(A \cap C) = P(A \cap B) = 1/4 = P(A)P(C)$, and similarly B and C are independent.

4. Give an example of 3 events A, B, C which are not independent, yet satisfy $P(A \cap B \cap C) = P(A)P(B)P(C)$. Hint: consider simple and extreme cases.

Consider the extreme case where $P(A) = 0$. Then it's automatically true that $P(A)P(B)P(C) = 0$, and also $P(A \cap B \cap C) = 0$ since $A \cap B \cap C$ is a subset of A (and in general, if event A_1 is a subset of event A_2 , then $P(A_1) \leq P(A_2)$). Then take B and C to be dependent, e.g., take any example with $B = C$ and $0 < P(B) < 1$.

1. A bag contains one marble which is either green or blue, with equal probabilities. A green marble is put in the bag (so there are 2 marbles now), and then a random marble is taken out. The marble taken out is green. What is the probability that the remaining marble is also green?

Historical note: this problem was first posed by Lewis Carroll in 1893.

Let A be the event that the initial marble is green, B be the event that the removed marble is green, and C be the event that the remaining marble is green. We need to find $P(C|B)$. There are several ways to find this; one natural way is to condition on whether the initial marble is green:

$$P(C|B) = P(C|B, A)P(A|B) + P(C|B, A^c)P(A^c|B) = 1P(A|B) + 0P(A^c|B).$$

To find $P(A|B)$, use Bayes' Rule:

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)} = \frac{1/2}{P(B|A)P(A) + P(B|A^c)P(A^c)} = \frac{1/2}{1/2 + 1/4} = \frac{2}{3}.$$

So $P(C|B) = 2/3$.

2. A spam filter is designed by looking at commonly occurring phrases in spam. Suppose that 80% of email is spam. In 10% of the spam emails, the phrase “free money” is used, whereas this phrase is only used in 1% of non-spam emails. A new email has just arrived, which does mention “free money”. What is the probability that it is spam?

Let S be the event that an email is spam and F be the event that an email has the “free money” phrase. By Bayes’ Rule,

$$P(S|F) = \frac{P(F|S)P(S)}{P(F)} = \frac{0.1 \cdot 0.8}{0.1 \cdot 0.8 + 0.01 \cdot 0.2} = \frac{80/1000}{82/1000} = \frac{80}{82} \approx 0.9756.$$

3. Let G be the event that a certain individual is guilty of a certain robbery. In gathering evidence, it is learned that an event E_1 occurred, and a little later it is also learned that another event E_2 also occurred.
- (a) Is it possible that individually, these pieces of evidence increase the chance of guilt (so $P(G|E_1) > P(G)$ and $P(G|E_2) > P(G)$), but together they decrease the chance of guilt (so $P(G|E_1, E_2) < P(G)$)?

Yes, this is possible. In fact, it is possible to have two events which separately provide evidence in favor of G , yet which together preclude G ! For example, suppose that the crime was committed between 1 pm and 3 pm on a certain day. Let E_1 be the event that the suspect was at a nearby coffeeshop from 1 pm to 2 pm that day, and let E_2 be the event that the suspect was at the nearby coffeeshop from 2 pm to 3 pm that day.

Then $P(G|E_1) > P(G)$, $P(G|E_2) > P(G)$ (assuming that being in the vicinity helps show that the suspect had the opportunity to commit the crime), yet $P(G|E_1 \cap E_2) < P(G)$ (as being in the coffeehouse from 1 pm to 3 pm gives the suspect an alibi for the full time).

(b) Show that the probability of guilt given the evidence is the same regardless of whether we update our probabilities all at once, or in two steps (after getting the first piece of evidence, and again after getting the second piece of evidence). That is, we can either update all at once (computing $P(G|E_1, E_2)$ in one step), or we can first update based on E_1 , so that our new probability function is $P_{\text{new}}(A) = P(A|E_1)$, and then update based on E_2 by computing $P_{\text{new}}(G|E_2)$.

This follows from the definition of conditional probability:

$$P_{\text{new}}(G|E_2) = \frac{P_{\text{new}}(G, E_2)}{P_{\text{new}}(E_2)} = \frac{P(G, E_2|E_1)}{P(E_2|E_1)} = \frac{P(G, E_1, E_2)/P(E_1)}{P(E_1, E_2)/P(E_1)} = P(G|E_1, E_2).$$

4. A crime is committed by one of two suspects, A and B . Initially, there is equal evidence against both of them. In further investigation at the crime scene, it is found that the guilty party had a blood type found in 10% of the population. Suspect A does match this blood type, whereas the blood type of Suspect B is unknown.

(a) Given this new information, what is the probability that A is the guilty party?

Let M be the event that A 's blood type matches the guilty party's and for brevity, write A for “ A is guilty” and B for “ B is guilty”. By Bayes' Rule,

$$P(A|M) = \frac{P(M|A)P(A)}{P(M|A)P(A) + P(M|B)P(B)} = \frac{1/2}{1/2 + (1/10)(1/2)} = \frac{10}{11}.$$

(We have $P(M|B) = 1/10$ since, given that B is guilty, the probability that A 's blood type matches the guilty party's is the same probability as for the general population.)

(b) Given this new information, what is the probability that B 's blood type matches that found at the crime scene?

Let C be the event that B 's blood type matches, and condition on whether B is guilty. This gives

$$P(C|M) = P(C|M, A)P(A|M) + P(C|M, B)P(B|M) = \frac{1}{10} \cdot \frac{10}{11} + \frac{1}{11} = \frac{2}{11}.$$

5. You are going to play 2 games of chess with an opponent whom you have never played against before (for the sake of this problem). Your opponent is equally likely to be a beginner, intermediate, or a master. Depending on which, your chances of winning an individual game are 90%, 50%, or 30%, respectively.

(a) What is your probability of winning the first game?

Let W_i be the event of winning the i th game. By the law of total probability,

$$P(W_1) = (0.9 + 0.5 + 0.3)/3 = 17/30.$$

(b) Congratulations: you won the first game! Given this information, what is the probability that you will also win the second game (assume that, given the skill level of your opponent, the outcomes of the games are independent)?

We have $P(W_2|W_1) = P(W_2, W_1)/P(W_1)$. The denominator is known from (a), while the numerator can be found by conditioning on the skill level of the opponent:

$$P(W_1, W_2) = \frac{1}{3}P(W_1, W_2|\text{beginner}) + \frac{1}{3}P(W_1, W_2|\text{intermediate}) + \frac{1}{3}P(W_1, W_2|\text{expert})$$

Since W_1 and W_2 are conditionally independent given the skill level of the opponent, this becomes

$$P(W_1, W_2) = (0.9^2 + 0.5^2 + 0.3^2)/3 = 23/60.$$

So

$$P(W_2|W_1) = \frac{23/60}{17/30} = 23/34.$$

(c) Explain the distinction between assuming that the outcomes of the games are independent and assuming that they are conditionally independent given the opponent's skill level. Which of these assumptions seems more reasonable, and why?

Independence here means that knowing one game's outcome gives no information about the other game's outcome, while conditional independence is the same statement where all probabilities are conditional on the opponent's skill level. Conditional independence given the opponent's skill level is a more reasonable assumption here. This is because winning the first game gives information about the opponent's skill level, which in turn gives information about the result of the second game.

That is, if the opponent's skill level is treated as fixed and known, then it may be reasonable to assume independence of games given this information; with the opponent's skill level random, earlier games can be used to help infer the opponent's skill level, which affects the probabilities for future games.

1. Consider the Monty Hall problem, except that Monty enjoys opening Door 2 more than he enjoys opening Door 3, and if he has a choice between opening these two doors, he opens Door 2 with probability p , where $\frac{1}{2} \leq p \leq 1$.

To recap: there are three doors, behind one of which there is a car (which you want), and behind the other two of which there are goats (which you don't want). Initially, all possibilities are equally likely for where the car is. You choose a door, which for concreteness we assume is Door 1. Monty Hall then opens a door to reveal a goat, and offers you the option of switching. Assume that Monty Hall knows which door has the car, will always open a goat door and offer the option of switching, and as above assume that if Monty Hall has a choice between opening Door 2 and Door 3, he chooses Door 2 with probability p (with $\frac{1}{2} \leq p \leq 1$).

- (a) Find the unconditional probability that the strategy of always switching succeeds (unconditional in the sense that we do not condition on which of Doors 2,3 Monty opens).

Let C_j be the event that the car is hidden behind door j and let W be the event that we win using the switching strategy. Using the law of total probability, we can find the unconditional probability of winning in the same way as in class:

$$\begin{aligned} P(W) &= P(W|C_1)P(C_1) + P(W|C_2)P(C_2) + P(W|C_3)P(C_3) \\ &= 0 \cdot 1/3 + 1 \cdot 1/3 + 1 \cdot 1/3 = 2/3. \end{aligned}$$

- (b) Find the probability that the strategy of always switching succeeds, given that Monty opens Door 2.

A tree method works well here (delete the paths which are no longer relevant after the conditioning, and reweight the remaining values by dividing by their sum), or we can use Bayes' rule and the law of total probability (as below).

Let D_i be the event that Monty opens Door i . Note that we are looking for $P(W|D_2)$, which is the same as $P(C_3|D_2)$ as we first choose Door 1 and then switch to Door 3. By Bayes' rule and the law of total probability,

$$\begin{aligned}
 P(C_3|D_2) &= \frac{P(D_2|C_3)P(C_3)}{P(D_2)} \\
 &= \frac{P(D_2|C_3)P(C_3)}{P(D_2|C_1)P(C_1) + P(D_2|C_2)P(C_2) + P(D_2|C_3)P(C_3)} \\
 &= \frac{1 \cdot 1/3}{p \cdot 1/3 + 0 \cdot 1/3 + 1 \cdot 1/3} \\
 &= \frac{1}{1+p}.
 \end{aligned}$$

(c) Find the probability that the strategy of always switching succeeds, given that Monty opens Door 3.

The structure of the problem is the same as part (b) (except for the condition that $p \geq 1/2$, which was not needed above). Imagine repainting doors 2 and 3, reversing which is called which. By part (b) with $1 - p$ in place of p , $P(C_2|D_3) = \frac{1}{1+(1-p)} = \frac{1}{2-p}$.

2. For each statement below, either show that it is true or give a counterexample. Throughout, X, Y, Z are discrete random variables.

(a) If X and Y are independent and Y and Z are independent, then X and Z are independent.

False: for a simple example, take $X = Z$.

(b) If X and Y are independent, then they are conditionally independent given Z .

False: this was discussed in class (the fire-popcorn example) in terms of events, for which we can let X, Y, Z be indicators.

(c) If X and Y are conditionally independent given Z , then they are independent.

False: this was discussed in class in terms of events (the chess opponent of unknown strength example); a coin with a random bias (as on HW 2) is another simple, useful example to keep in mind.

(d) If X and Y have the same distribution given Z , i.e., for all a and z , we have $P(X = a|Z = z) = P(Y = a|Z = z)$, then X and Y have the same distribution.

True: by the law of total probability, conditioning on Z gives

$$P(X = a) = \sum_z P(X = a|Z = z)P(Z = z).$$

Since X and Y have the same conditional distribution given Z , this becomes $\sum_z P(Y = a|Z = z)P(Z = z) = P(Y = a)$.

1. (a) Is it possible to have events A, B, E such that $P(A|E) < P(B|E)$ and $P(A|E^c) < P(B|E^c)$, yet $P(A) > P(B)$? That is, A is less likely under B given that E is true, and also given that E is false, yet A is more likely than B if given no information about E . Show this is impossible (with a short proof) or find a counterexample (with a “story” interpreting A, B, E).

It is *not* possible, as seen using the law of total probability:

$$P(A) = P(A|E)P(E) + P(A|E^c)P(E^c) < P(B|E)P(E) + P(B|E^c)P(E^c) = P(B).$$

- (b) Is it possible to have events A, B, E such that $P(A|B, E) < P(A|B^c, E)$ and $P(A|B, E^c) < P(A|B^c, E^c)$, yet $P(A|B) > P(A|B^c)$? That is, given that E is true, learning B is evidence against A , and similarly given that E is false; but given no information about E , learning that B is true is evidence in favor of A . Show this is impossible (with a short proof) or find a counterexample (with a “story” interpreting A, B, E).

Yes, this is possible: this is the structure of Simpson’s Paradox! For example, consider the Stampy problem above. Or, consider the two doctors example discussed in class: suppose that there are two doctors, Dr. Hibbert and Dr. Nick. Each performs two types of surgery, say heart transplants and bandaid removals. Let A be the event that a surgery is successful, let B be the event that Dr. Nick performs the surgery and B^c be the complement, that Dr. Hibbert performs the surgery. Let E be heart surgery and E^c be bandaid removal.

Is it possible that Dr. Hibbert is better than Dr. Nick at both heart transplants and bandaid removals, yet Dr. Nick has a higher success rate overall? Yes, this can happen if Dr. Nick performs mostly bandaid removals and Dr. Hibbert performs mostly heart transplants. That is, the better doctor may perform relatively more of the harder surgery, resulting in a lower success rate.

To make up specific numbers, suppose that Dr. Hibbert performed 90 heart transplants, with 70 successful, and 10 bandaid removals, with all 10 successful. Dr. Nick performed 10 heart transplants, with 2 successful, and 90 bandaid removals, with 81 successful. Formally, our probability space consists of randomly choosing one of the 200 surgeries, uniformly. Note that each individual surgery is more likely to be successful given that it's performed by Dr. Hibbert, but Dr. Hibbert's overall success rate (80%) is lower than Dr. Nick's (83%).

There are many real-life examples of Simpson's Paradox. For example, it is possible for one baseball player to have a higher batting average than another in each of two seasons, yet a lower batting average when the two seasons are aggregated. Simpson's Paradox illustrates the importance of controlling for additional variables that interfere with the analysis (known as *confounders*).

2. Consider the following conversation from an episode of *The Simpsons*:

Lisa: *Dad, I think he's an ivory dealer! His boots are ivory, his hat is ivory, and I'm pretty sure that check is ivory.*

Homer: *Lisa, a guy who's got lots of ivory is less likely to hurt Stampy than a guy whose ivory supplies are low.*

Here Homer and Lisa are debating the question of whether or not the man (named Blackheart) is likely to hurt Stampy the Elephant if they sell Stampy to him. They clearly disagree about how to use their observations about Blackheart to learn about the probability (conditional on the evidence) that Blackheart will hurt Stampy.

- (a) Define clear notation for the various events of interest here.

Let H be the event that the man will hurt Stampy, let L be the event that a man has lots of ivory, and let D be the event that the man is an ivory dealer.

- (b) Express Lisa's and Homer's arguments (Lisa's is partly implicit) as conditional probability statements in terms of your notation from (a).

Lisa observes that L is true. She suggests (reasonably) that this evidence makes D more likely, i.e., $P(D|L) > P(D)$. Implicitly, she suggests that this makes it likely that the man will hurt Stampy, i.e.,

$$P(H|L) > P(H|L^c).$$

Homer argues that

$$P(H|L) < P(H|L^c).$$

(c) Assume it is true that someone who has a lot of a commodity will have less desire to acquire more of the commodity. Explain what is wrong with Homer's reasoning that the evidence about Blackheart makes it less likely that he will harm Stampy.

Homer does not realize that observing that Blackheart has so much ivory makes it much more likely that Blackheart is an ivory dealer, which in turn makes it more likely that the man will hurt Stampy. (This is an example of Simpson's Paradox.) It may be true that, *controlling for whether or not Blackheart is a dealer*, having high ivory supplies makes it less likely that he will harm Stampy: $P(H|L, D) < P(H|L^c, D)$ and $P(H|L, D^c) < P(H|L^c, D^c)$. However, this does not imply that $P(H|L) < P(H|L^c)$.

1. A gambler repeatedly plays a game where in each round, he wins a dollar with probability $1/3$ and loses a dollar with probability $2/3$. His strategy is "quit when he is ahead by \$2," though some suspect he is a gambling addict anyway. Suppose that he starts with a million dollars. Show that the probability that he'll ever be ahead by \$2 is less than $1/4$.

This problem is a special case of the gambler's ruin. Let A_1 be the event that he is successful on the first play and let W be the event that he is ever ahead by \$2 before being ruined. Then by the law of total probability, we have

$$P(W) = P(W|A_1)P(A_1) + P(W|A_1^c)P(A_1^c).$$

Let a_i be the probability that the gambler achieves a profit of \$2 before being ruined, starting with a fortune of $\$i$. For our setup, $P(W) = a_i$, $P(W|A_1) = a_{i+1}$ and $P(W|A_1^c) = a_{i-1}$. Therefore,

$$a_i = a_{i+1}/3 + 2a_{i-1}/3,$$

with boundary conditions $a_0 = 0$ and $a_{i+2} = 1$. We can then solve this difference equation for a_i (directly or using the result of the gambler's ruin problem):

$$a_i = \frac{2^i - 1}{2^{2+i} - 1}.$$

This is always less than $1/4$ since $\frac{2^i - 1}{2^{2+i} - 1} < \frac{1}{4}$ is equivalent to $4(2^i - 1) < 2^{2+i} - 1$, which is equivalent to the true statement $2^{2+i} - 4 < 2^{2+i} - 1$.

3. A family has 3 children, creatively named A , B , and C .

(a) Discuss intuitively (but clearly) whether the event “ A is older than B ” is independent of the event “ A is older than C .”

They are not independent: knowing that A is older than B makes it more likely that A is older than C , as the if A is older than B , then the only way that A can be younger than C is if the birth order is CAB , whereas the birth orders ABC and ACB are both compatible with A being older than B . To make this more intuitive, think of an extreme case where there are 100 children instead of 3, call them $A_1, \dots, A_{100}$. Given that A_1 is older than all of $A_2, A_3, \dots, A_{99}$, it's clear that A_1 is very old (relatively), whereas there isn't evidence about where A_{100} fits into the birth order.

(b) Find the probability that A is older than B , given that A is older than C .

Writing $x > y$ to mean that x is older than y ,

$$P(A > B | A > C) = \frac{P(A > B, A > C)}{P(A > C)} = \frac{1/3}{1/2} = \frac{2}{3}$$

since $P(A > B, A > C) = P(A \text{ is the eldest child}) = 1/3$ (unconditionally, any of the 3 children is equally likely to be the eldest).

4. Two coins are in a hat. The coins look alike, but one coin is fair (with probability $1/2$ of Heads), while the other coin is biased, with probability $1/4$ of Heads. One of the coins is randomly pulled from the hat, without knowing which of the two it is. Call the chosen coin “Coin C”.

(a) Coin C is tossed twice, showing Heads both times. Given this information, what is the probability that Coin C is the fair coin?

By Bayes’ Rule,

$$P(\text{fair}|HH) = \frac{P(HH|\text{fair})P(\text{fair})}{P(HH)} = \frac{(1/4)(1/2)}{(1/4)(1/2) + (1/16)(1/2)} = \frac{4}{5}.$$

(b) Are the events “first toss of Coin C is Heads” and “second toss of Coin C is Heads” independent? Explain.

They’re not independent: the first toss being Heads is evidence in favor of the coin being the fair coin, giving information about probabilities for the second toss.

(c) Find the probability that in 10 flips of Coin C, there will be exactly 3 Heads. (The coin is equally likely to be either of the 2 coins; do not assume it already landed Heads twice as in (a).)

Let X be the number of Heads in 10 tosses. By the Law of Total Probability (conditioning on which of the two coins C is),

$$\begin{aligned} P(X = 3) &= P(X = 3|\text{fair})P(\text{fair}) + P(X = 3|\text{biased})P(\text{biased}) \\ &= \binom{10}{3}(1/2)^{10}(1/2) + \binom{10}{3}(1/4)^3(3/4)^7(1/2) \\ &= \frac{1}{2} \binom{10}{3} \left(\frac{1}{2^{10}} + \frac{3^7}{4^{10}} \right). \end{aligned}$$

5. A woman has been murdered, and her husband is accused of having committed the murder. It is known that the man abused his wife repeatedly in the past, and the prosecution argues that this is important evidence pointing towards the man's guilt. The defense attorney says that the history of abuse is irrelevant, as only 1 in 1000 men who beat their wives end up murdering them.

Assume that the defense attorney's 1 in 1000 figure is correct, and that half of men who murder their wives previously abused them. Also assume that 20% of murdered women were killed by their husbands, and that if a woman is murdered and the husband is *not* guilty, then there is only a 10% chance that the husband abused her. What is the probability that the man is guilty? Is the prosecution right that the abuse is important evidence in favor of guilt?

The defense attorney claims that the probability of a man murdering his wife given that he abuses her is 1/1000. Note though that this conditional probability is not what is in issue. We need to *condition on all the evidence*, and here the evidence is not only that the man beat his wife, but also that the wife did get murdered. That is, we need the probability of guilt given that the husband abused his wife *and* that she was murdered. Let's introduce some notation to specify the different events.

G : the man is guilty of murdering his wife

M : the wife was murdered

A : the man abused his wife

The relevant probability is $P(G|A, M)$. We are given that $P(A|G, M) = 0.5$, $P(G|M) = 0.2$, $P(A|G^c, M) = 0.1$.

$$P(G|A, M) = \frac{P(A|G, M)P(G|M)}{P(A|M)} = \frac{P(A|G, M)P(G|M)}{P(A|G, M)P(G|M) + P(A|G^c, M)P(G^c|M)}$$

(This is Bayes' Rule, where the probabilities are all taken as conditional M .) So

$$P(G|A, M) = \frac{0.5 \cdot 0.2}{0.5 \cdot 0.2 + 0.1 \cdot 0.8} = \frac{5}{9}.$$

This means that the evidence of abuse raised the probability of guilt from 20% to 56%, certainly important evidence. Since the woman was indeed murdered, it is crucial to condition on that information when weighing other evidence.

6. A family has two children. Assume that birth month is independent of gender, with boys and girls equally likely and all months equally likely, and assume that the elder child's characteristics are independent of the younger child's characteristics).

(a) Find the probability that both are girls, given that the elder child is a girl who was born in March.

Let G_j be the event that the j th born child is a girl and M_j be the event that the j th born child was born in March, for $j \in \{1, 2\}$. Then $P(G_1 \cap G_2 | G_1 \cap M_1) = P(G_2 | G_1 \cap M_1)$, since if we know that G_1 occurs, then $G_1 \cap G_2$ occurring is the same thing as G_2 occurring. By independence of the characteristics of the children, $P(G_2 | G_1 \cap M_1) = P(G_2) = 1/2$.

(b) Find the probability that both are girls, given that at least one is a girl who was born in March.

$$\begin{aligned} P(\text{both girls} | \text{at least one March-born girl}) &= \frac{P(\text{both girls, at least one born in March})}{P(\text{at least one March-born girl})} \\ &= \frac{(1/4)(1 - (11/12)^2)}{1 - (23/24)^2} \\ &= \frac{23}{47} \\ &\approx 0.489. \end{aligned}$$

In contrast, $P(\text{both girls} | \text{at least one girl}) = 1/3$. So the seemingly irrelevant “born in March” information actually matters! By symmetry, the answer would stay the same if we replaced “born in March” by, say, “born in July”; so it's not the awesomeness of March that matters, but rather the fact that the information brings “at least one” closer to being “a specific one.” The more detailed the information being conditioned on, the closer this becomes to specifying one of the children, and thus the closer the answer gets to $1/2$.

1. (a) Consider the following 7-door version of the Monty Hall problem. There are 7 doors, behind one of which there is a car (which you want), and behind the rest of which there are goats (which you don't want). Initially, all possibilities are equally likely for where the car is. You choose a door. Monty Hall then opens 3 goat doors, and offers you the option of switching to any of the remaining 3 doors.

Assume that Monty Hall knows which door has the car, will always open 3 goat doors and offer the option of switching, and that Monty chooses with equal probabilities from all his choices of which goat doors to open. Should you switch? What is your probability of success if you switch to one of the remaining 3 doors?

Assume the doors are labeled such that you choose Door 1 (to simplify notation), and suppose first that you follow the “stick to your original choice” strategy. Let S be the event of success in getting the car, and let C_j be the event that the car is behind Door j . Conditioning on which door has the car, we have

$$P(S) = P(S|C_1)P(C_1) + \cdots + P(S|C_7)P(C_7) = P(C_1) = \frac{1}{7}.$$

Let M_{ijk} be the event that Monty opens Doors i, j, k . Then

$$P(S) = \sum_{i,j,k} P(S|M_{ijk})P(M_{ijk})$$

(summed over all i, j, k with $2 \leq i < j < k \leq 7$.) By symmetry, this gives

$$P(S|M_{ijk}) = P(S) = \frac{1}{7}$$

for all i, j, k with $2 \leq i < j < k \leq 7$. Thus, the conditional probability that the car is behind 1 of the remaining 3 doors is $6/7$, which gives $2/7$ for each. So you should switch, thus making your probability of success $2/7$ rather than $1/7$.

(b) Generalize the above to a Monty Hall problem where there are $n \geq 3$ doors, of which Monty opens m goat doors, with $1 \leq m \leq n - 2$.

By the same reasoning, the probability of success for “stick to your original choice” is $\frac{1}{n}$, both unconditionally and conditionally. Each of the $n - m - 1$ remaining doors has conditional probability $\frac{n-1}{(n-m-1)n}$ of having the car. This value is greater than $\frac{1}{n}$, so you should switch, thus obtaining probability $\frac{n-1}{(n-m-1)n}$ of success (both conditionally and unconditionally).

2. The *odds* of an event with probability p are defined to be $\frac{p}{1-p}$, e.g., an event with probability $3/4$ is said to have odds of 3 to 1 in favor (or 1 to 3 against). We are interested in a hypothesis H (which we think of as a event), and we gather new data as evidence (expressed as an event D) to study the hypothesis. The *prior* probability of H is our probability for H being true before we gather the new data; the *posterior* probability of H is our probability for it after we gather the new data. The *likelihood ratio* is defined as $\frac{P(D|H)}{P(D|H^c)}$.

(a) Show that Bayes' rule can be expressed in terms of odds as follows: *the posterior odds of a hypothesis H are the prior odds of H times the likelihood ratio.*

We want to show that

$$\frac{P(H|D)}{P(H^c|D)} = \frac{P(H)}{P(H^c)} \frac{P(D|H)}{P(D|H^c)}$$

By Bayes' rule, we have

$$\begin{aligned} P(H|D) &= P(D|H)P(H)/P(D), \\ P(H^c|D) &= P(D|H^c)P(H^c)/P(D). \end{aligned}$$

Dividing the first of these by the second, we immediately obtain the desired equation, which is a useful alternative way to write Bayes' rule.

(b) As in the example from class, suppose that a patient tests positive for a disease afflicting 1% of the population. For a patient who has the disease, there is a 95% chance of testing positive (in medical statistics, this is called the *sensitivity* of the test); for a patient who doesn't have the disease, there is a 95% chance of testing negative test (in medical statistics, this is called the *specificity* of the test).

The patient gets a second, independent test done (with the same sensitivity and specificity), and again tests positive. Use the odds form of Bayes' rule to find the probability that the patient has the disease, given the evidence, *in two ways*: in one step, conditioning on both test results simultaneously, and in two steps, first updating the probabilities based on the first test result, and then updating again based on the second test result.

To go from odds back to probability, we divide odds by (1 plus odds), since $\frac{p/q}{1+p/q} = p$ for $q = 1 - p$. Let H be the event of having the disease. The prior odds are 99 to 1 against having the disease. The likelihood ratio based on one test result is $\frac{0.95}{0.05}$. Now we can immediately carry out either the one update or the two update method.

One update method: The likelihood ratio based on both test results is $\frac{0.95^2}{0.05^2}$ since the tests are independent. So the posterior odds of the patient having the disease are

$$\frac{1}{99} \cdot \frac{0.95^2}{0.05^2} = \frac{361}{99} \approx 3.646,$$

which corresponds to a probability of $361/(361 + 99) = 361/460 \approx 0.78$ (whereas based on one positive test, the probability was only 0.16 of having the disease).

Two updates method: After the first test, the posterior odds of the patient having the disease are

$$\frac{1}{99} \cdot \frac{0.95}{0.05} \approx 0.19,$$

which corresponds to a probability of $0.19/(1+0.19) \approx 0.16$ (agreeing with the result from class). These posterior odds become the new prior odds, and then updating based on the second test gives $(\frac{1}{99} \cdot \frac{0.95}{0.05}) \cdot \frac{0.95}{0.05}$, which is the same result as above.

3. Is it possible to have events A_1, A_2, B, C with $P(A_1|B) > P(A_1|C)$ and $P(A_2|B) > P(A_2|C)$, yet $P(A_1 \cup A_2|B) < P(A_1 \cup A_2|C)$? If so, find an example (with a “story” interpreting the events, as well as giving specific numbers); otherwise, show that it is impossible for this phenomenon to happen.

Yes, this is possible. First note that $P(A_1 \cup A_2|B) = P(A_1|B) + P(A_2|B) - P(A_1 \cap A_2|B)$, so it is *not* possible if A_1 and A_2 are disjoint, and that it is crucial to consider the intersection. So let’s choose examples where $P(A_1 \cap A_2|B)$ is much larger than $P(A_1 \cap A_2|C)$, to offset the other inequalities.

Story 1: Consider two basketball players, one of whom is randomly chosen to shoot two free throws. The first player is very streaky, and always either makes both or misses both free throws, with probability 0.8 of making both (this is an extreme example chosen for simplicity, but we could also make it so the player has good days (on which there is a high chance of making both shots) and bad days (on which there is a high chance of missing both shots) without requiring *always* making both or missing both). The second player’s free throws go in with probability 0.7, independently. Define the events as A_j : the j th free throw goes in; B : the free throw shooter is the first player; $C = B^c$. Then

$$P(A_1|B) = P(A_2|B) = P(A_1 \cap A_2|B) = P(A_1 \cup A_2|B) = 0.8,$$

$$P(A_1|C) = P(A_2|C) = 0.7, P(A_1 \cap A_2|C) = 0.49, P(A_1 \cup A_2|C) = 2 \cdot 0.7 - 0.49 = 0.91.$$

Story 2: Suppose that you can either take Good Class or Other Class, but not both. If you take Good Class, you’ll attend lecture 70% of the time, and you will understand the material if and only if you attend lecture. If you take Other Class, you’ll attend lecture 40% of the time and understand the material 40% of the time, but because the class is so poorly taught, the only way you understand the material is by studying on your own and not attending lecture. Defining the events as A_1 : attend lecture; A_2 : understand material; B : take Good Class; C : take Other Class,

$$P(A_1|B) = P(A_2|B) = P(A_1 \cap A_2|B) = P(A_1 \cup A_2|B) = 0.7,$$

$$P(A_1|C) = P(A_2|C) = 0.4, P(A_1 \cap A_2|C) = 0, P(A_1 \cup A_2|C) = 2 \cdot 0.4 = 0.8.$$

4. Calvin and Hobbes play a match consisting of a series of games, where Calvin has probability p of winning each game (independently). They play with a “win by two” rule: the first player to win two games more than his opponent wins the match. Find the probability that Calvin wins the match (in terms of p), in two different ways:

(a) by conditioning, using the law of total probability.

Let C be the event that Calvin wins the match, $X \sim \text{Bin}(2, p)$ be how many of the first 2 games he wins, and $q = 1 - p$. Then

$$P(C) = P(C|X = 0)q^2 + P(C|X = 1)(2pq) + P(C|X = 2)p^2 = 2pqP(C) + p^2,$$

so $P(C) = \frac{p^2}{1-2pq}$. This can also be written as $\frac{p^2}{p^2+q^2}$, since $p + q = 1$.

Miracle check: Note that this should (and does) reduce to 1 for $p = 1$, 0 for $p = 0$, and $\frac{1}{2}$ for $p = \frac{1}{2}$. Also, it makes sense that the probability of Hobbes winning, which is $1 - P(C) = \frac{q^2}{p^2+q^2}$, can also be obtained by swapping p and q .

(b) by interpreting the problem as a gambler’s ruin problem.

The problem can be thought of as a gambler’s ruin where each player starts out with \$2. So the probability that Calvin wins the match is

$$\frac{1 - (q/p)^2}{1 - (q/p)^4} = \frac{(p^2 - q^2)/p^2}{(p^4 - q^4)/p^4} = \frac{(p^2 - q^2)/p^2}{(p^2 - q^2)(p^2 + q^2)/p^4} = \frac{p^2}{p^2 + q^2},$$

which agrees with the above.

5. A fair die is rolled repeatedly, and a running total is kept (which is, at each time, the total of all the rolls up until that time). Let p_n be the probability that the running total is ever *exactly* n (assume the die will always be rolled enough times so that the running total will eventually exceed n , but it may or may not ever equal n).

(a) Write down a recursive equation for p_n (relating p_n to earlier terms p_k in a simple way). Your equation should be true for all positive integers n , so give a definition of p_0 and p_k for $k < 0$ so that the recursive equation is true for small values of n .

We will find something to condition on to reduce the case of interest to earlier, simpler cases. This is achieved by the useful strategy of *first step analysis*. Let p_n be the probability that the running total is ever *exactly* n . Note that if, for example, the first throw is a 3, then the probability of reaching n exactly is p_{n-3} since starting from that point, we need to get a total of $n - 3$ exactly. So

$$p_n = \frac{1}{6}(p_{n-1} + p_{n-2} + p_{n-3} + p_{n-4} + p_{n-5} + p_{n-6}),$$

where we define $p_0 = 1$ (which makes sense anyway since the running total is 0 before the first toss) and $p_k = 0$ for $k < 0$.

(b) Find p_7 .

Using the recursive equation in (a), we have

$$\begin{aligned} p_1 &= \frac{1}{6}, \quad p_2 = \frac{1}{6}\left(1 + \frac{1}{6}\right), \quad p_3 = \frac{1}{6}\left(1 + \frac{1}{6}\right)^2, \\ p_4 &= \frac{1}{6}\left(1 + \frac{1}{6}\right)^3, \quad p_5 = \frac{1}{6}\left(1 + \frac{1}{6}\right)^4, \quad p_6 = \frac{1}{6}\left(1 + \frac{1}{6}\right)^5. \end{aligned}$$

Hence,

$$p_7 = \frac{1}{6}(p_1 + p_2 + p_3 + p_4 + p_5 + p_6) = \frac{1}{6} \left(\left(1 + \frac{1}{6}\right)^6 - 1 \right) \approx 0.2536.$$

(c) Give an intuitive explanation for the fact that $p_n \rightarrow 1/3.5 = 2/7$ as $n \rightarrow \infty$.

An intuitive explanation is as follows. The average number thrown by the die is (total of dots)/6, which is $21/6 = 7/2$, so that every throw adds on an average of $7/2$. We can therefore expect to land on 2 out of every 7 numbers, and the probability of landing on any particular number is $2/7$. This result can be proved as follows (a proof was *not* required):

$$\begin{aligned} & p_{n+1} + 2p_{n+2} + 3p_{n+3} + 4p_{n+4} + 5p_{n+5} + 6p_{n+6} \\ &= p_{n+1} + 2p_{n+2} + 3p_{n+3} + 4p_{n+4} + 5p_{n+5} \\ &\quad + p_n + p_{n+1} + p_{n+2} + p_{n+3} + p_{n+4} + p_{n+5} \\ &= p_n + 2p_{n+1} + 3p_{n+2} + 4p_{n+3} + 5p_{n+4} + 6p_{n+5} \\ &= \dots \\ &= p_{-5} + 2p_{-4} + 3p_{-3} + 4p_{-2} + 5p_{-1} + 6p_0 = 6. \end{aligned}$$

Taking the limit of the lefthand side as n goes to ∞ , we have

$$(1 + 2 + 3 + 4 + 5 + 6) \lim_{n \rightarrow \infty} p_n = 6,$$

so $\lim_{n \rightarrow \infty} p_n = 2/7$.